

Responsible AI Impact Assessment

Malaria Cell Classification A Computer-Vision Screening Aid Prototype

1. Who Does This System Affect?

This system is a computer-vision prototype intended to support malaria-cell image classification in chronically understaffed Level 2 and Level 3 public health facilities across malaria-endemic counties in western Kenya particularly Homa Bay, Kisumu, Migori, and Siaya. Its primary affected users are laboratory and health workers, who might use an automated prediction as an additional screening aid; patients are affected indirectly, insofar as model outputs could influence whether a smear is flagged as potentially parasitized.

The model was trained and evaluated on Bangladeshi clinical samples and has not been validated on Kenyan-collected blood smears. Differences in staining protocols, microscope equipment, image quality, and local clinical practice could cause real-world performance in Kenya to diverge from the reported test-set results. The system is accordingly positioned as an educational and research prototype, not a clinical diagnostic device.

2. What Is the Worst Plausible Outcome?

The most serious plausible outcome is a false-negative prediction on a parasitized cell: a patient whose smear contains malaria parasites could receive an “Uninfected” classification. This risk is concrete rather than hypothetical, the model's recall on the Parasitized class is 91.29%, meaning roughly 1 in 11 parasitized cells in the source-domain test set is currently missed. In a real clinical workflow, inappropriate reliance on such an output could contribute to delayed confirmation, delayed treatment, or delayed referral.

This risk is amplified by the fact that the model has only been evaluated on source-domain data and has not yet been validated on Kenyan smears; true performance in the target deployment context is currently unknown. Grad-CAM visualizations are included to show which image regions contributed to a prediction, but highlighting a region is not the same as confirming it is medically meaningful or that the prediction itself is correct. For all of these reasons, the system must never replace microscopy, laboratory confirmation, or qualified clinical judgement.

3. Who Bears the Cost of Errors?

The greatest cost of a false-negative error falls on the patient, who may experience a delayed or missed diagnosis if an automated prediction is incorrectly trusted in place of confirmatory testing. Health workers also bear costs: an inaccurate prediction can create additional verification work, or gradually undermine confidence in the system as a whole. False-positive predictions carry a different but real cost, unnecessary review or confirmatory testing that adds to an already strained workload.

Because the training data originate from Bangladesh, Kenyan patients and health workers bear an additional, structural risk from dataset shift: strong performance on the source dataset does not establish equivalent performance on Kenyan clinical images, and this gap will not be visible until it is explicitly measured.

4. What Redress Mechanism Exists?

The system should operate strictly as a decision-support or research tool, with mandatory human review — never as an autonomous diagnostic system. Every prediction should be independently checked against established clinical and laboratory procedures before it informs any medical decision.

- Where a prediction appears incorrect, the image and the corresponding laboratory determination should be logged for error analysis and future model improvement.
- Users should have a clear, low-friction mechanism to report incorrect predictions, poor-quality images, or other deployment failures.
- No patient should be denied testing, treatment, referral, or further investigation solely on the basis of this model's prediction.

5. How Will the Model Be Monitored?

Before any clinical deployment, the model should be evaluated on Kenyan-collected smears, followed by continuous evaluation using appropriately governed validation data. Performance should be tracked using sensitivity/recall, specificity, precision, F1-score, confusion matrices, and ROC-AUC — never accuracy alone, which can mask weak minority-class performance under class imbalance.

Monitoring should be disaggregated by relevant data characteristics such as collection site, staining protocol, microscope type, and image-quality category, where those attributes are available and ethical to collect. Any substantial performance degradation should trigger investigation and suspension of automated use until the cause is understood, and model updates should be versioned, independently evaluated, and compared against the existing model before deployment.

Fairness Criterion: Equal Opportunity

Equal Opportunity is the most appropriate of the four standard fairness criteria for this system, because the principal safety concern is whether the model reliably detects the clinically important positive class (Parasitized cells). Equal Opportunity focuses specifically on equality of true-positive rates across relevant groups. In a Kenyan deployment, this means checking whether parasitized-cell detection remains comparably sensitive across different collection sites, staining protocols, microscope equipment, or other relevant subgroups, rather than only reporting an aggregate recall figure.

Demographic Parity is inappropriate here because the underlying proportion of parasitized cells should not necessarily be equal across populations, enforcing equal prediction rates would fight the true underlying distribution rather than accuracy. Individual Fairness is difficult to operationalize without a principled definition of which images should be treated as “similar.” Equalized Odds is broader, requiring equality of both true-positive and false-positive rates, but given the immediate safety priority of not missing parasitized cells, equalizing true-positive performance is the criterion that matches the actual risk.

Disaggregated Findings

Class	Precision	Recall	F1-score	Support
Uninfected	0.9185	0.9816	0.9490	2,067
Parasitized	0.9803	0.9129	0.9454	2,067
Macro average	0.9494	0.9473	0.9472	4,134

Overall discrimination: ROC-AUC = 0.9860, indicating strong separability between the Parasitized and Uninfected classes at the aggregate, source-domain level.

Against the project's 90% recall screening threshold, no evaluated class currently falls below the bar. However, the Parasitized class shows substantially lower recall (91.29%) than the Uninfected class (98.16%) — a ~7-point gap that makes missed parasitized cells the single most important current performance risk. This asymmetry is precisely why an aggregate ROC-AUC of 0.986 should not be read as evidence of clinical readiness: a high overall discrimination score is compatible with a clinically material minority-class miss rate, and it says nothing about performance in the Kenyan target domain, where no evaluation has yet been conducted.

Datasheet Summary

The dataset represents Bangladeshi clinical blood-smear samples used to train and evaluate the classifier. It does not represent Kenyan-collected smears from Homa Bay, Kisumu, Migori, or Siaya, and therefore does not establish how the model performs across Kenyan staining practices, microscopes, image-quality conditions, or local clinical workflows. Any deployment decision should treat this as an open validation gap, not a formality, the assessment above is only complete once Kenyan-domain performance has been measured against the same disaggregated criteria used here.